

Lab Report 06: Wireless Routing

Experiment Name: Wireless Routing Configuration and Connectivity

Introduction

The objective of this lab is to configure a wireless local area network (WLAN) using a Linksys WRT300N Wireless Router and verify connectivity between wired and wireless end-devices. The lab focuses on the router's ability to bridge different physical mediums (Ethernet and Radio Frequency) while managing IP distribution via DHCP.

Tools

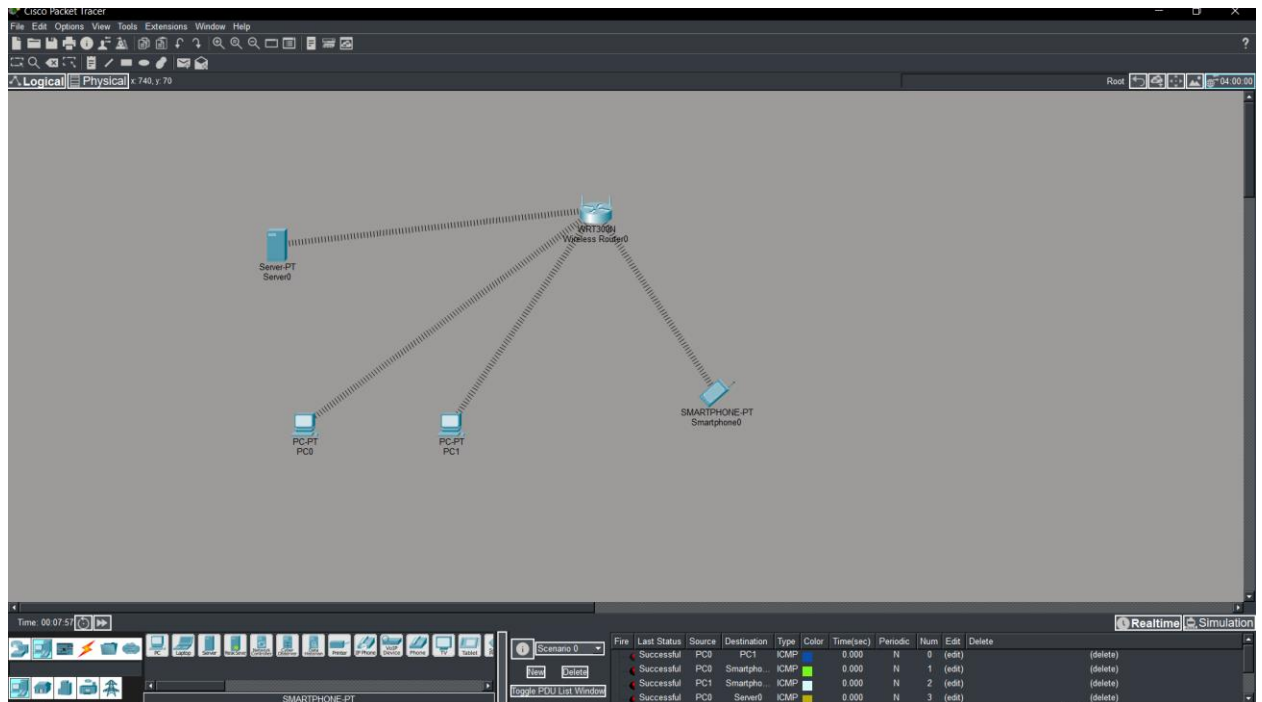
- **Software:** Cisco Packet Tracer
 - **Devices:** * 1 x Wireless Router (WRT300N)
 - 1 x Server-PT (Server0)
 - 2 x PC-PT (PC0, PC1)
 - 1 x Smartphone-PT (Smartphone0)
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Procedure

1. Network Topology Construction:

- The WRT300N Wireless Router was placed as the central gateway.
- **Server0**, **PC0**, and **PC1** were connected via Copper Straight-Through cables to the router's Ethernet LAN ports.
- **Smartphone0** was configured with a wireless interface to connect to the router's SSID.

2.



Wireless Configuration:

- Accessed the Router's **GUI** tab to verify the **SSID** and **Wireless Channel**.
- Ensured the **DHCP Server** was enabled to provide automated IP addressing to all connected clients.

Wireless Router0
Physical
Config
GUI
Attributes

Wireless-N Broadband Router
Firmware Version: v0.93.3

Setup
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Administration
Status

Basic Setup
DDNS
MAC Address Clone
Advanced Routing

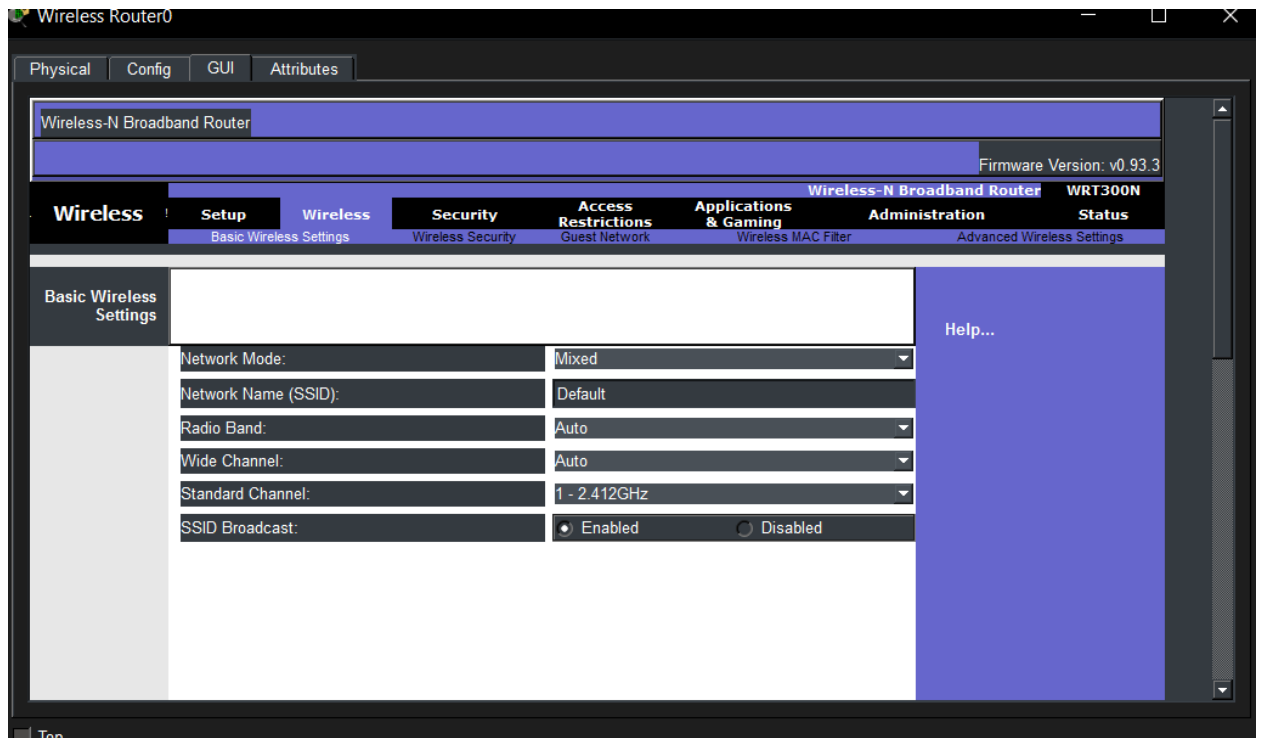
Internet Setup
Internet Connection type
Optional Settings (required by some internet service providers)
Network Setup
Router IP
DHCP Server Settings

Automatic Configuration - DHCP
Host Name:
Domain Name:
MTU:
Size: 1500
IP Address:
192
168
0
1
Subnet Mask:
255.255.255.0
DHCP Server:
Enabled
Disabled
DHCP Reservation
Start IP Address:
192.168.0
100
Maximum number of Users:
50
IP Address Range:
192.168.0
100
149
Client Lease Time:
0
minutes (0 means one day)
Static DNS 1:
0
0
0
0
Static DNS 2:
0
0
0
0
Static DNS 3:
0
0
0
0
WINS:
0
0
0
0

Help...

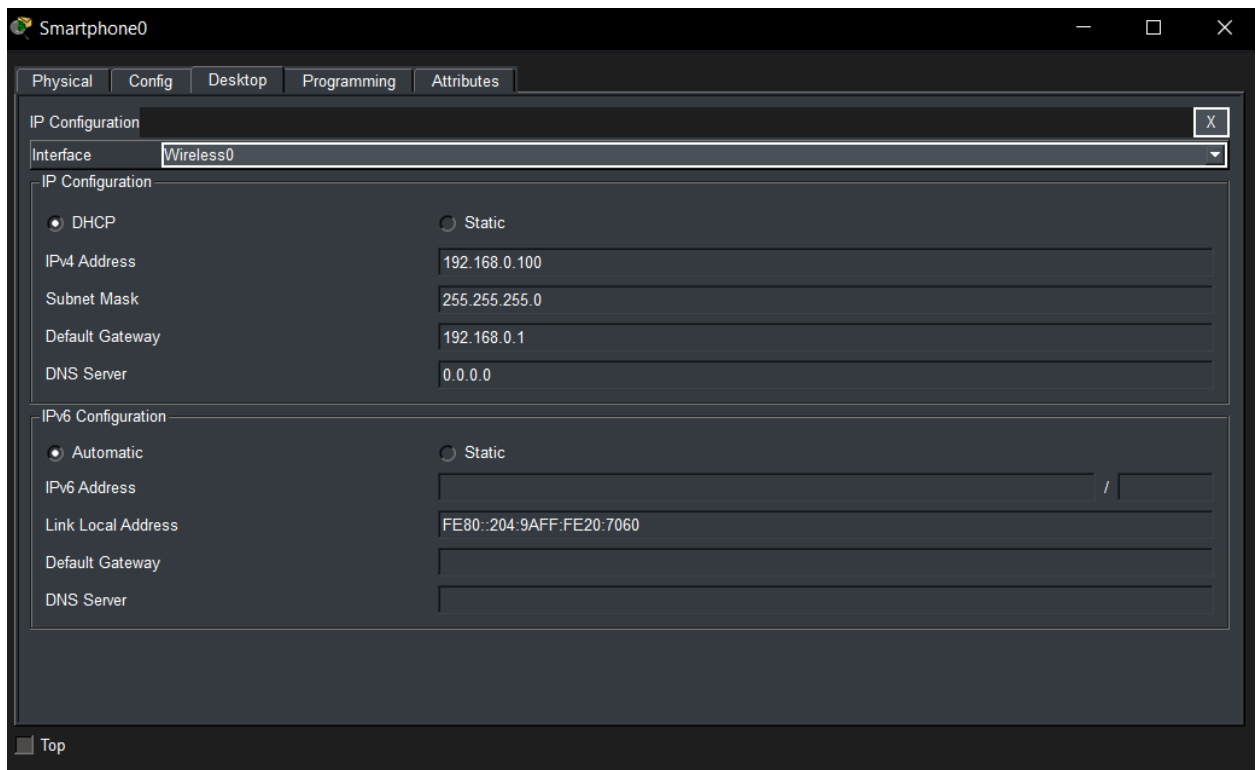
Top

3.



4. IP Address Verification:

- Checked the **IP Configuration** on **Smartphone0** and **PC0** to ensure they received valid IP addresses within the same subnet.



4.

5. Connectivity Testing:

- Executed ICMP Echo Requests (Pings) between the wired PCs, the Server, and the Wireless Smartphone.
- The **PDU List Window** was used to monitor the status of these packets.

Realtime Simulation										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
✓	Successful	PC0	PC1	ICMP	Blue	0.000	N	0	(edit)	(delete)
✓	Successful	PC0	Smartpho...	ICMP	Green	0.000	N	1	(edit)	(delete)
✓	Successful	PC1	Smartpho...	ICMP	Cyan	0.000	N	2	(edit)	(delete)
✓	Successful	PC0	Server0	ICMP	Yellow	0.000	N	3	(edit)	(delete)

Conclusion

This experiment successfully demonstrated the functionality of a wireless router in a mixed-media environment. By configuring the **WRT300N**, we established a network where wired devices (PCs/Server) could communicate seamlessly with wireless devices (Smartphone).

The success of the lab was verified by the **PDU List**, which showed "Successful" status for all cross-device pings. This confirms that the router effectively handled the encapsulation of data across different layers and that the **DHCP protocol** correctly managed the addressing scheme, ensuring a fully reachable and functional network.

